

Project Report

Objective:

This project implements a smart attendance system using RFID cards and a microcontroller to streamline and automate the process of tracking attendance. The system displays user identification data on an LCD screen and logs attendance information to an SD card.

Theory of Operation:

This system uses an MFRC522 RFID module to read unique ID cards. When an authorized tag is scanned, the Arduino Mega compares the scanned UID against a predefined list of authorized RFID card identifiers and fetches the associated user name. The information is then displayed on a 16x2 LCD screen, and a log (name + timestamp) is written to an 8GB micro SD card.

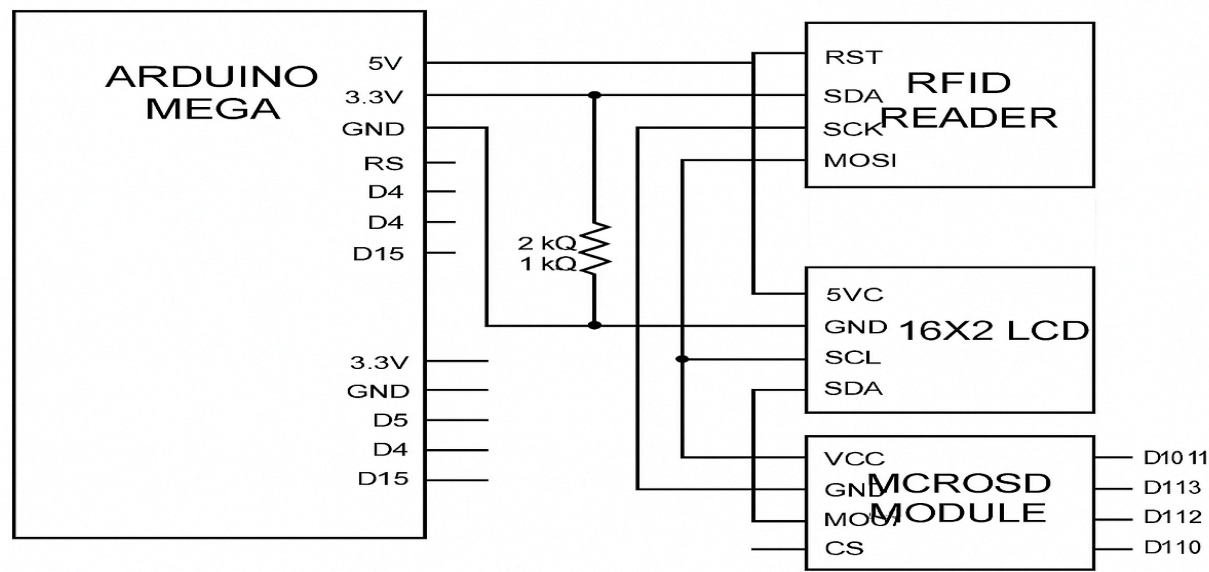
System workflow:

RFID Scan → Arduino Processing → Display Name & Access status → Save to SD Card

Equipment:

Component	Model	Documentation Link
RFID Module	MFRC522	Datasheet
Arduino	Mega 2560	Datasheet
LCD Display	16x2 with I2C	Datasheet
SD Card Module	SPI-Compatible	Tutorial
SD Card	8GB, Class 4	Specs
Jumper Wires	Male-to-Male	Overview
LED + Resistor	5mm + 220Ω	Datasheet

Schematic:



Procedure:

Step	Description
1	Assemble components on a breadboard according to schematic
2	Upload the Arduino code to Mega 2560 using USB
3	Test RFID module with authorized RFID cards
4	On successful scan, LCD displays name & time
5	System logs attendance to micro SD card
6	Repeat for multiple users and validate log entries

Communication Interfaces Used:

SPI:

- MFRC522 RFID Reader
- SD Card Module

I2C:

16x2 LCD Display

Serial:

- Arduino Serial Monitor for debugging and validation

The project required coordinating multiple communication interfaces while maintaining reliable RFID detection, LCD updates, and SD card logging operations.

Learning Outcomes and Project Complexity

This project provided us with valuable hands-on experience in building modular embedded systems using Arduino. We structured our code into four distinct files, each responsible for a specific function, which improved maintainability and made the system easier to test and expand.

keycard_test.ino — Main Runtime Program

This is the core of the system, handling real-time operations. It performs several critical tasks:

Tasks performed:

- RFID Scanning: Detects and reads RFID tags using the MFRC522 library.
- UID Matching: Compares the scanned UID against a predefined list of authorized RFID card identifiers.
- Display Handling: Shows the user's name, time, and access status on the LCD.
- Data Logging: Records each scan to the SD card in CSV format.

- Feedback Loop: Outputs logs and errors through the Serial Monitor for debugging.

read_file.ino — Attendance Log Reader

This utility program reads attendance records stored on the micro SD card and outputs the contents through the Serial Monitor for validation and troubleshooting.

Tasks performed:

- Opens the attendance log file from the SD card
- Reads stored attendance entries
- Displays log contents through the Serial Monitor
- Assists with validation and debugging of attendance records

del_file.ino — Attendance Log Reset Utility

This utility program removes the attendance log file from the micro SD card, allowing the system to be reset and retested without previous attendance records.

Tasks performed:

- Locates the attendance log file
- Deletes existing attendance records
- Supports repeated testing and validation cycles
- Simplifies system maintenance during development

set_name.ino — RFID Card Programming Utility

This utility program writes user identification information to RFID card memory for testing and validation purposes.

Tasks performed:

- Programs user information onto RFID cards
- Supports RFID card preparation and testing
- Verifies successful card-writing operations
- Assists with system validation during development

Technical Challenges

One significant technical challenge involved integrating both the MFRC522 RFID reader and SD card module on the same SPI bus. Because both peripherals shared SPI communication lines, proper chip-select management was required to prevent bus contention and communication failures. Through debugging and iterative testing, the system was modified to reliably switch between RFID transactions and SD card logging operations while maintaining stable device communication.

We also had to account for voltage compatibility. The RFID module operates at 3.3V logic, while the Arduino Mega outputs 5V. To prevent damage, we used a voltage divider made from two resistors to safely step down the signal levels. This taught us about logic level conversion and the importance of protecting low-voltage devices.

We relied heavily on the Serial Monitor throughout development. It was essential for debugging issues such as RFID detection failures, SD card write failures, SPI communication

conflicts, and UID mismatches. Iterative testing and debugging helped improve the reliability and stability of the final system.

Key Skills Developed

- RFID-based access control systems
- Embedded C/C++ development
- Serial communication for user interaction and debugging
- Modular code design and hardware abstraction
- Multi-device SPI coordination and signal buffering
- Voltage level shifting using resistors
- Real-time data logging and feedback systems

Overall Assessment

Overall, the project was moderately to highly challenging. It gave us experience with real-world constraints in embedded design and prepared us for more advanced systems involving IoT, automation, hardware communication, and embedded system integration.

